

Integral
polytopes

A polytope is **integral** if all its vertices are in $\mathbb{Z}^n$.
(lattice)

Every integral polytope can be written as

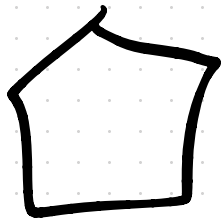
$$P = \{ \underbrace{\langle y_i, x \rangle}_{\text{inner product}} \leq b_i : \forall i \}$$

where $y_i \in \mathbb{Z}^n$, $b_i \in \mathbb{Z}$.

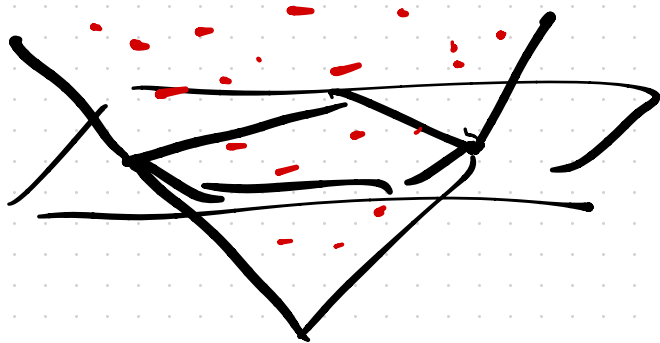
Converse is not true, any polytope of the above form is a **rational** polytope.

Given a polytope $P \subset \mathbb{R}^n$, define ^{"cone"} $C(P)$ to be

$$C(P) = \{ (tx, t) \in \mathbb{R}^{n+1} : t \geq 0 \}$$



$\mathbb{R}^n$



$\mathbb{R}^{n+1}$

Let $x_1, x_2, \dots, x_n, y$ be indeterminates. Let $A(P)$ be the k -vector space spanned by monomials of the form

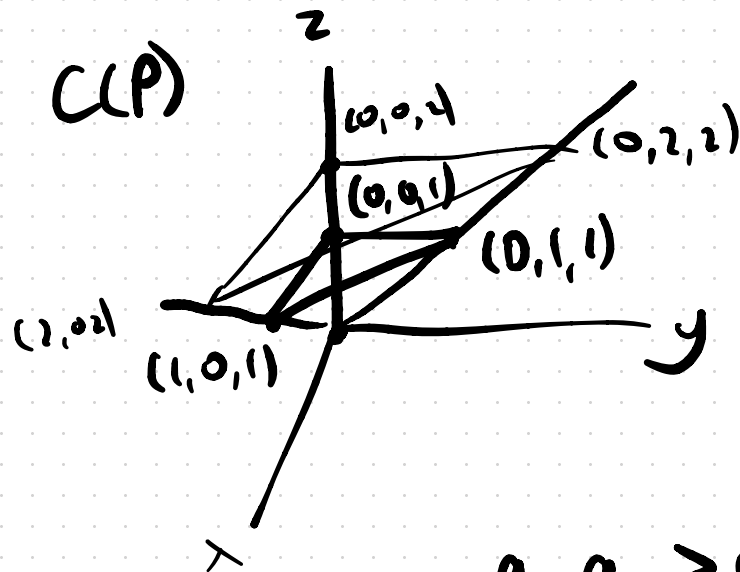
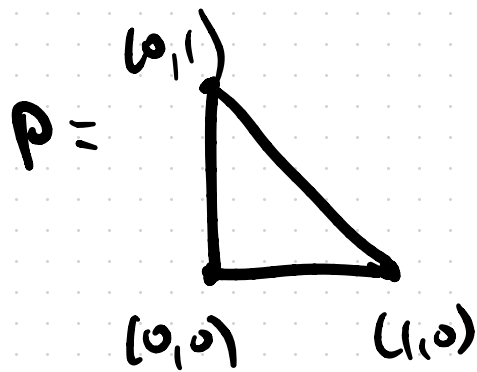
$$x_1^{a_1} x_2^{a_2} \dots x_n^{a_n} y^b$$

where $(a_1, \dots, a_n, b) \in C(P) \cap \mathbb{Z}^{n+1}$.

We define multiplication in $A(P)$ as expected.

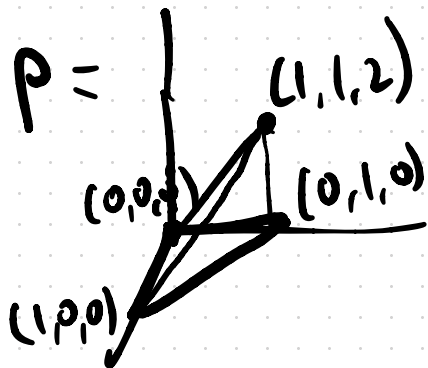
This makes $A(P)$ a graded k -algebra, with

$$\deg(x_1^{a_1} \dots x_n^{a_n} y^b) = b$$



$A(P)$ spanned by $x_1^{a_1} x_2^{a_2} y^b$ where $a_1 + a_2 \leq b$

As an algebra, $A(P)$ is generated by $y, x_1 y, x_2 y$



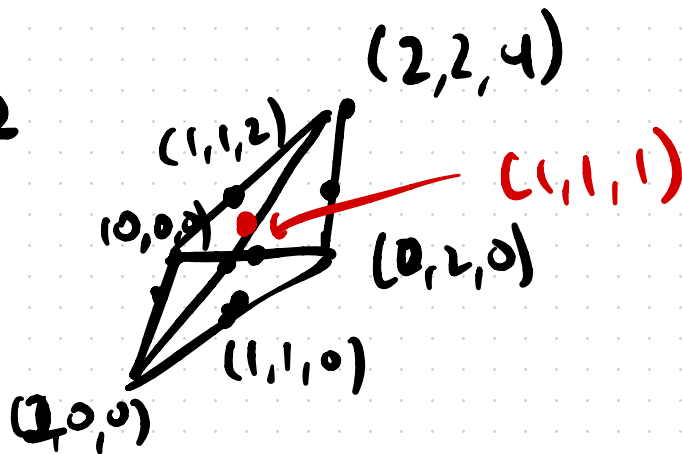
$$C(P) \subset \mathbb{R}^4$$

$$x_u = 0$$

$$x_u = 1$$

$$x_u = 2$$

$(0,0,0)$
vertices of P



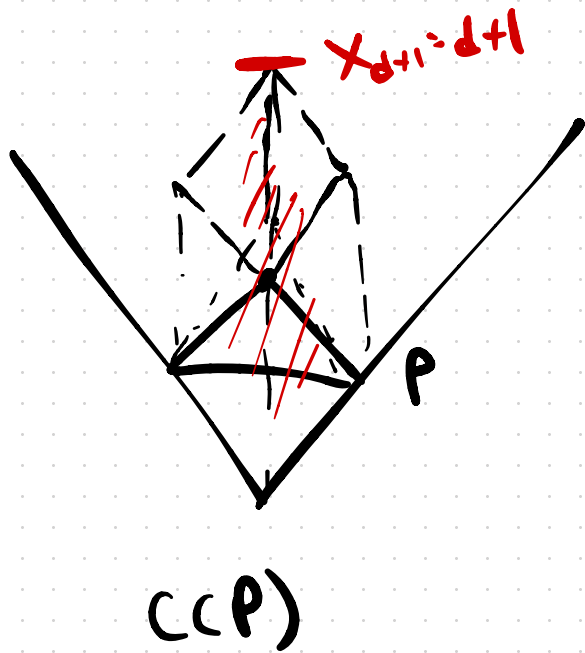
$(1,1,1)$ cannot be written as
a sum of integer points in P .

So in $A(P)$, the monomial $x_1 x_2 x_3 y^2$ is not
generated by degree 1 elements.

$A(P)$ generated by $y, x_1 y, x_2 y, x_1 x_2 x_3^2 y,$
 $x_1 x_2 x_3 y^2$

Prop: $A(P)$ is finitely generated.

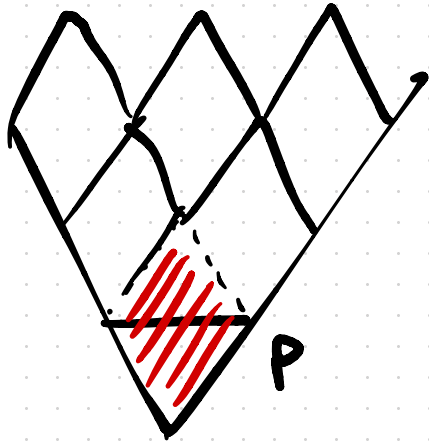
~~Proof: First assume P is a simplex.~~



Let $v_1, \dots, v_{d+1}$ be the vertices of P .

Let

$$D = \left\{ \sum_{i=1}^{d+1} r_i (v_i, 1) : 0 \leq r_i < 1 \right\}$$



Claim: Every integer point in $C(P)$ can be written as

$$\alpha + \sum_{i=1}^{d+1} q_i (v_i, 1)$$

where $\alpha \in D \cap \mathbb{Z}^{n+1}$, $q_i \in \mathbb{Z}_{\geq 0}$

Proof: Let $x \in C(P) \cap \mathbb{Z}^{n+1}$. Since $(v_i, 1)$ generate $C(P)$ as cone, we can write

$$x = \sum_{i=1}^{d+1} s_i (v_i, 1) \quad s_i \in \mathbb{Q}_{\geq 0}$$

$$\begin{aligned}
 x &= \sum_{i=1}^{d+1} (Ls_i + \underbrace{\{s_i\}}_{\text{fractional part}}) (v_i, 1) \quad 0 \leq \{s_i\} < 1 \\
 &= \sum_{i=1}^{d+1} \underbrace{Ls_i}_{q_i} (v_i, 1) + \underbrace{\sum_{i=1}^{d+1} \{s_i\} (v_i, 1)}_{D \cap \mathbb{Z}^{n+1}}
 \end{aligned}$$

If P is a simplex, the $(v_i, 1)$ are linearly independent, so this representation is unique.

ALP is not generated in degree 1.

In general, if A is a graded k -algebra generated by homogeneous elements of degrees $d_1, \dots, d_s$, then

$$H(A, t) = \frac{p(t)}{\prod (1 - t^{d_i})} \quad \text{for some polynomial } p.$$

Thm:

$$H(A(P), t) = \frac{h^*(t)}{(1-t)^{d+1}}$$

$$d = \dim P$$

h^* is a polynomial

" h^* -polynomial of P "

Proof: First assume P is a simplex.

As in the previous proof, every $x \in C(P) \cap \mathbb{Z}^{n+1}$ can be written uniquely as

$$\underbrace{\alpha}_{\in D \cap \mathbb{Z}^{n+1}} + \sum_{i=1}^{d+1} q_i (v_i, 1) \quad q_i \in \mathbb{Z}_{\geq 0}$$

Thus

$$H(\mathbb{K}[\underline{x}, \underline{y} : i=1, \dots, d+1], t)$$

$$H(A(p), t) = \sum_{\alpha \in D \cap \mathbb{Z}^{n+1}} t^{\deg \alpha} \cdot \frac{1}{(1-t)^{d+1}}$$

$$= \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

where h_i^* is the number of integer points in D at height i .